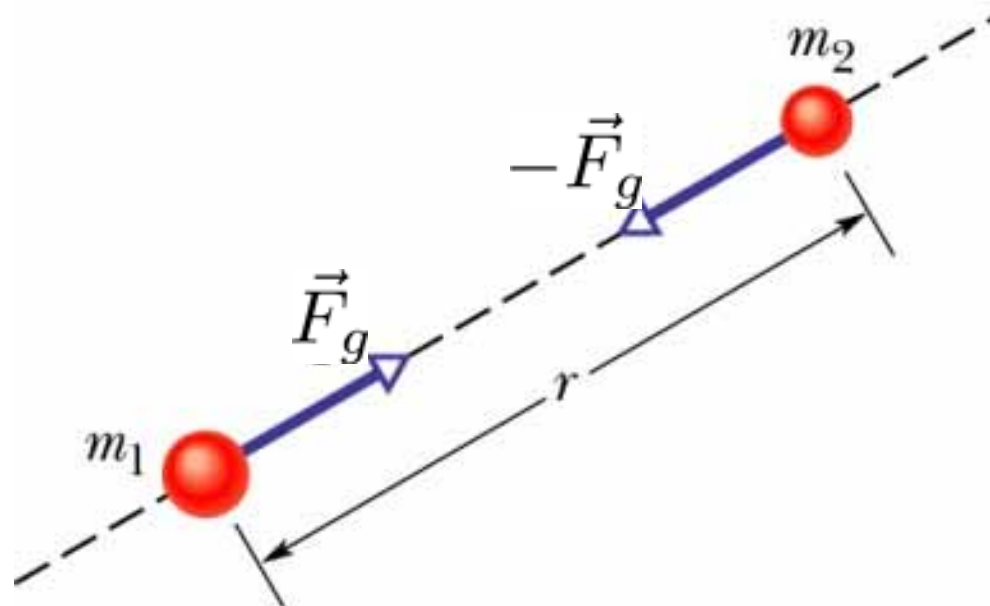


Elektrizität und Magnetismus

Kapitel 1: Das Elektron und andere Ladungen

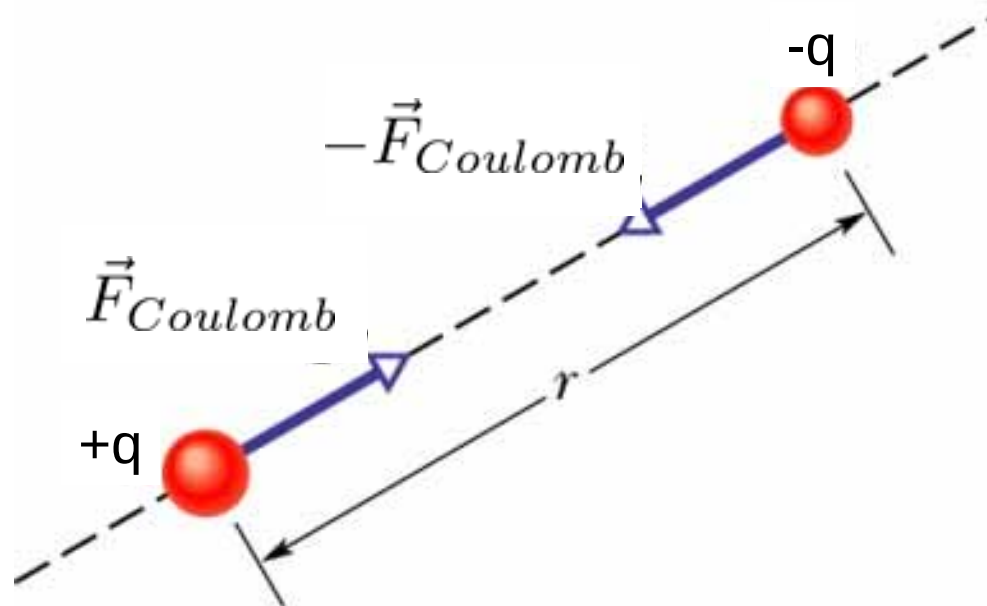
Prof. Axel Görlitz
Heinrich-Heine Universität Düsseldorf
Axel.Goerlitz@uni-duesseldorf.de

Die Gravitationskraft



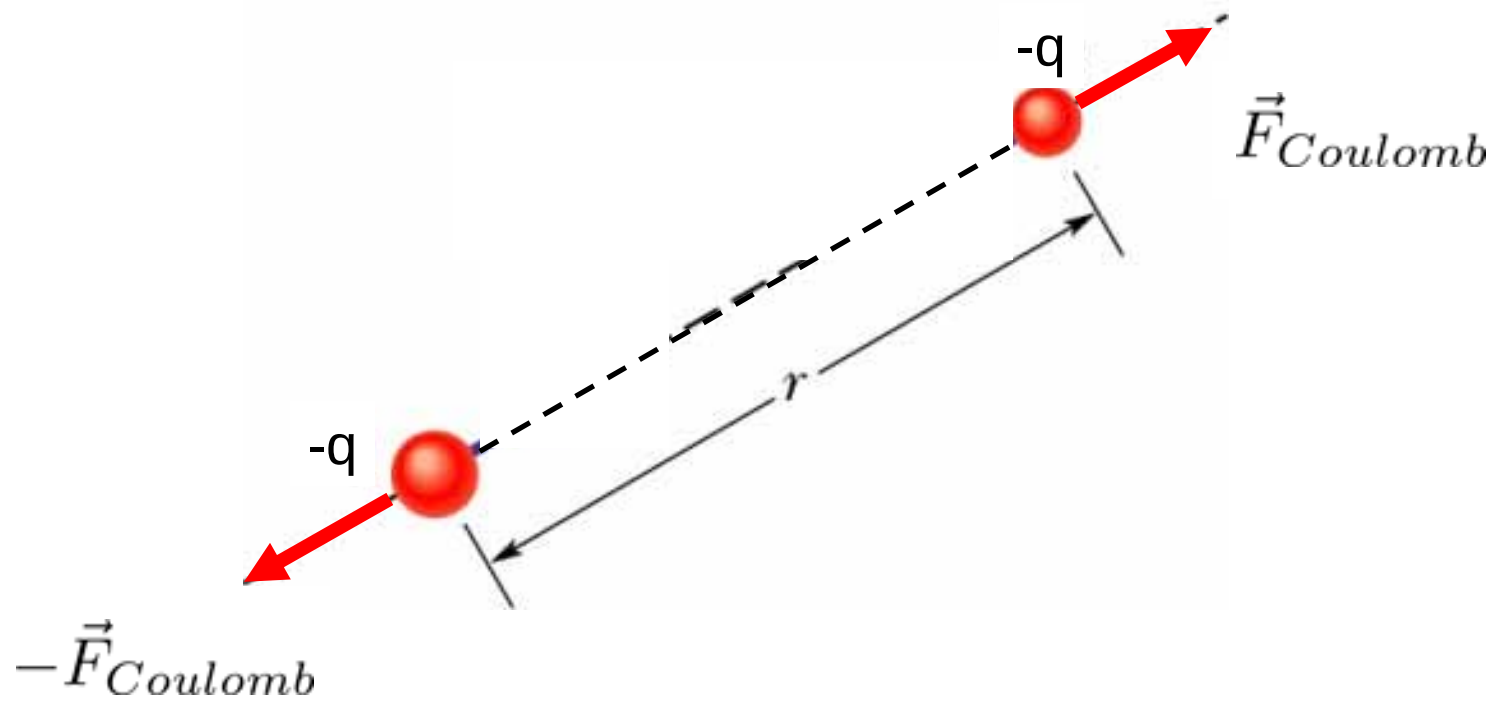
Halliday-Resnick-Walker (2003) Abb. 14.2

Die Coulombkraft



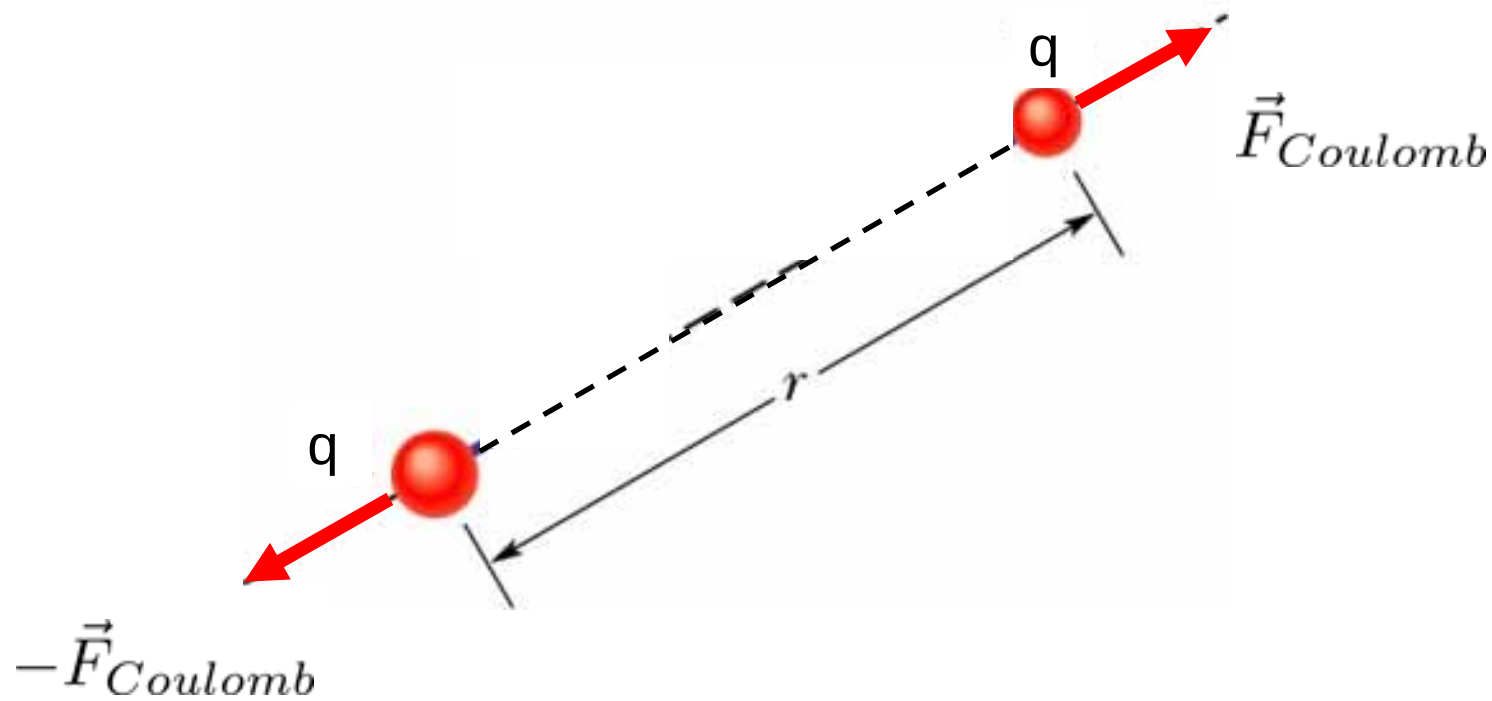
Halliday-Resnick-Walker (2003) Abb. 14.2

Die Coulombkraft



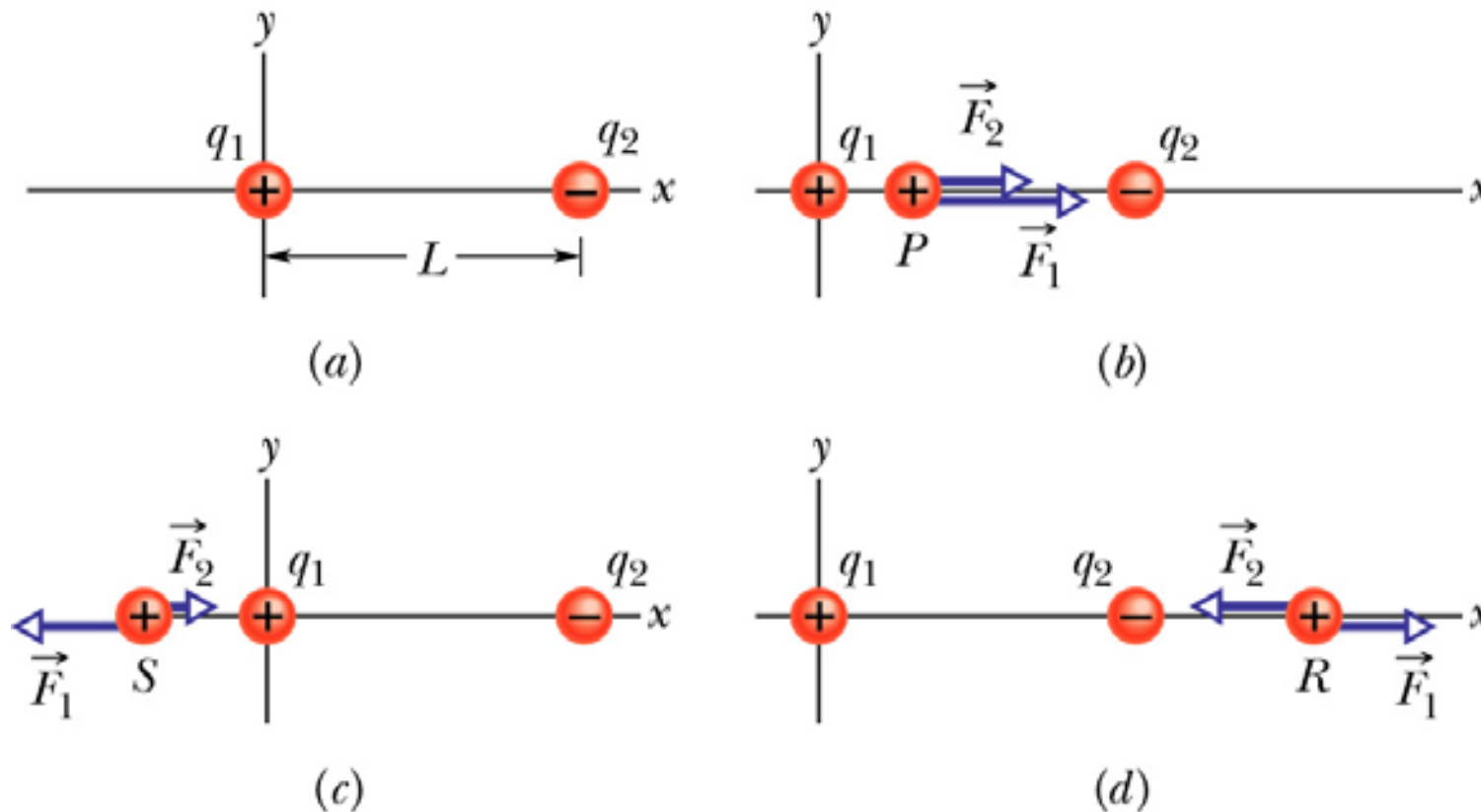
Halliday-Resnick-Walker (2003) Abb. 14.2

Die Coulombkraft



Halliday-Resnick-Walker (2003) Abb. 14.2

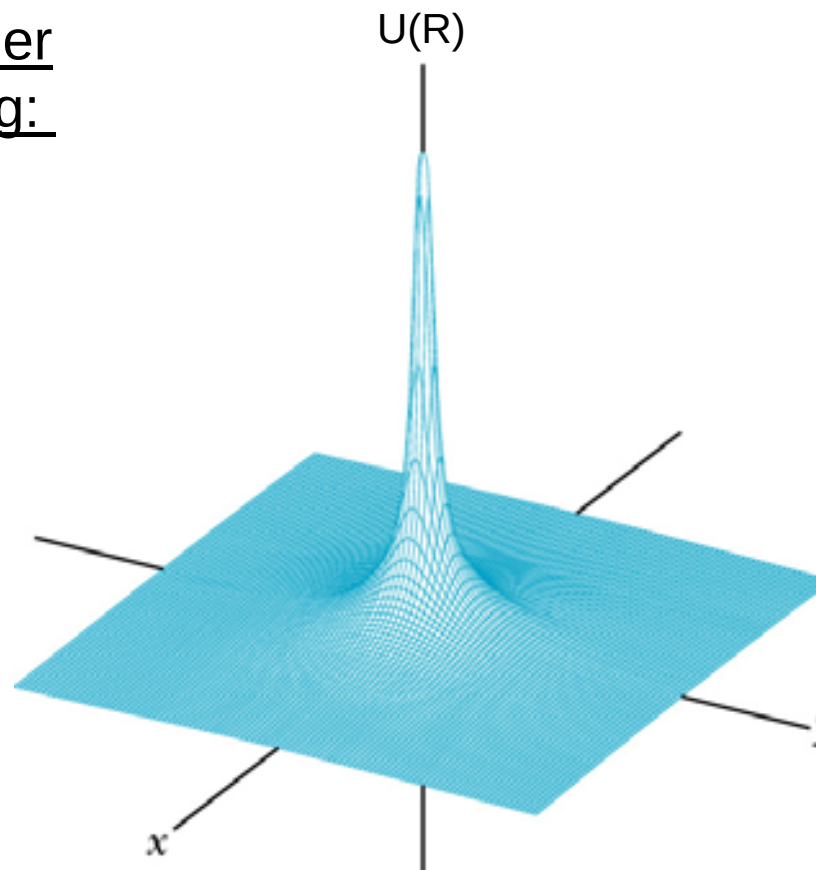
Die Coulombkraft



Halliday-Resnick-Walker (2003) Abb. 28.2

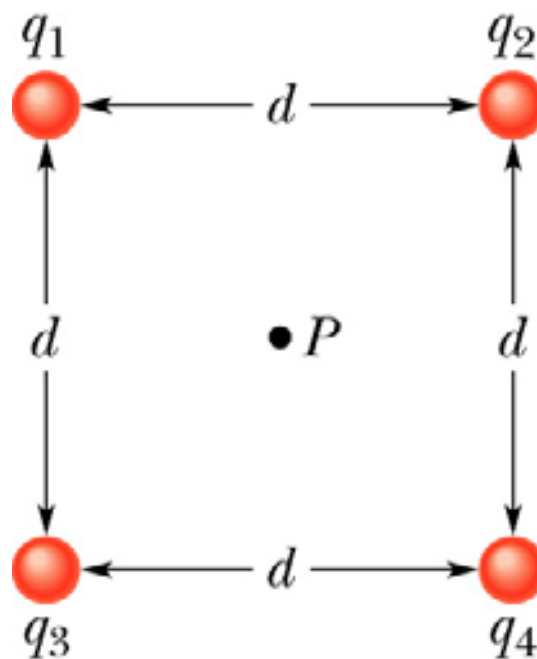
Coulombpotential

Coulombpotential einer
positiven Punktladung:



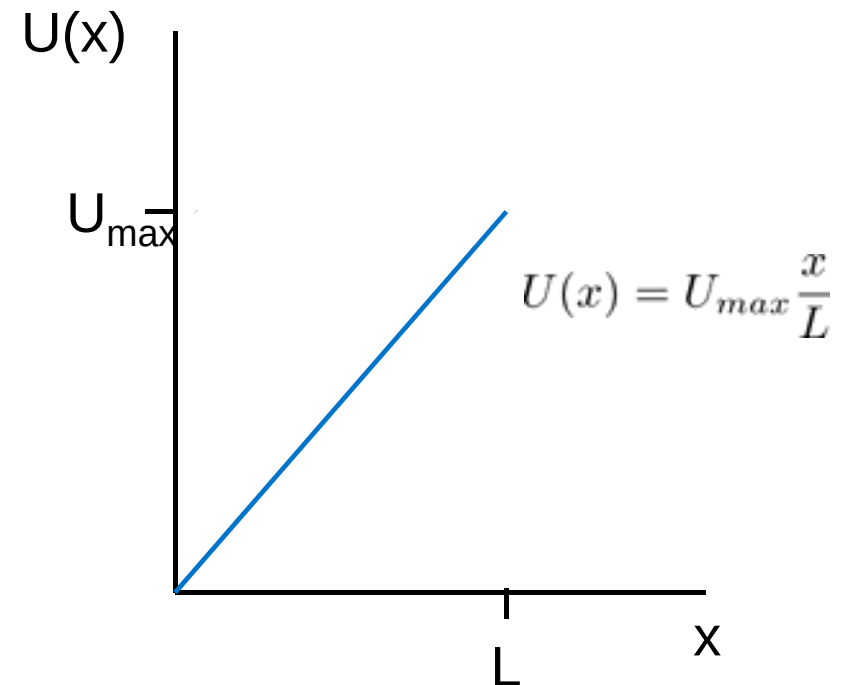
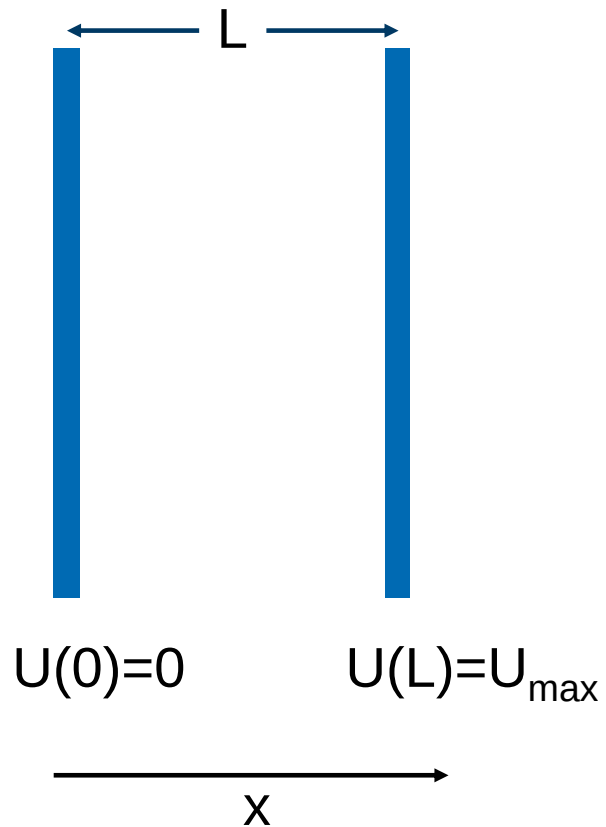
Halliday-Resnick-Walker (2003) Abb. 25.8

Elektrisches Potential

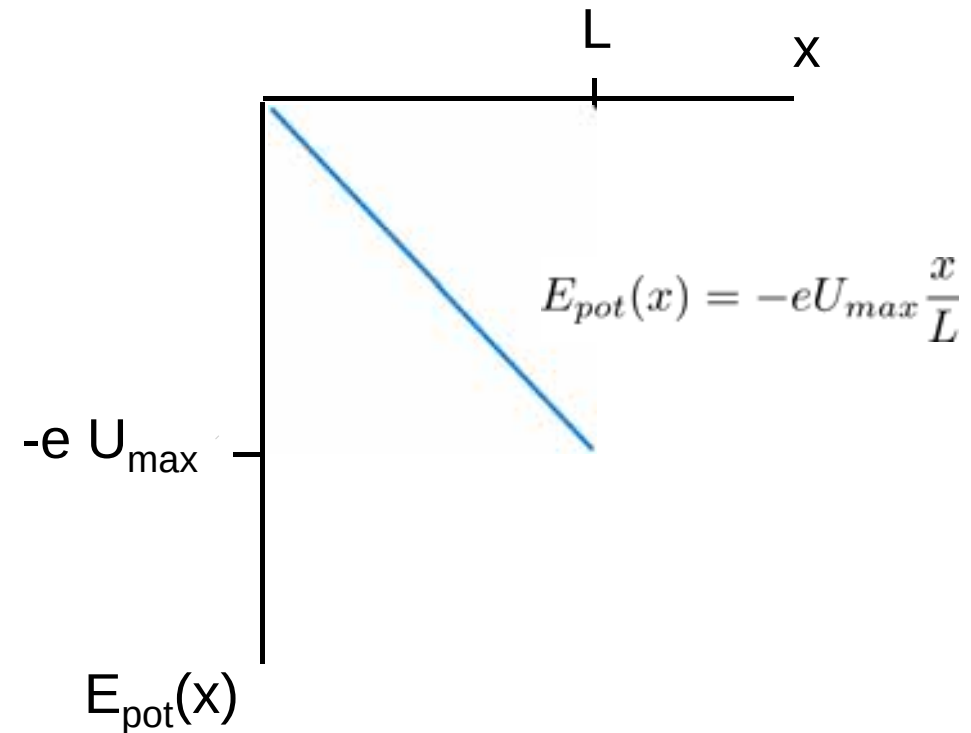
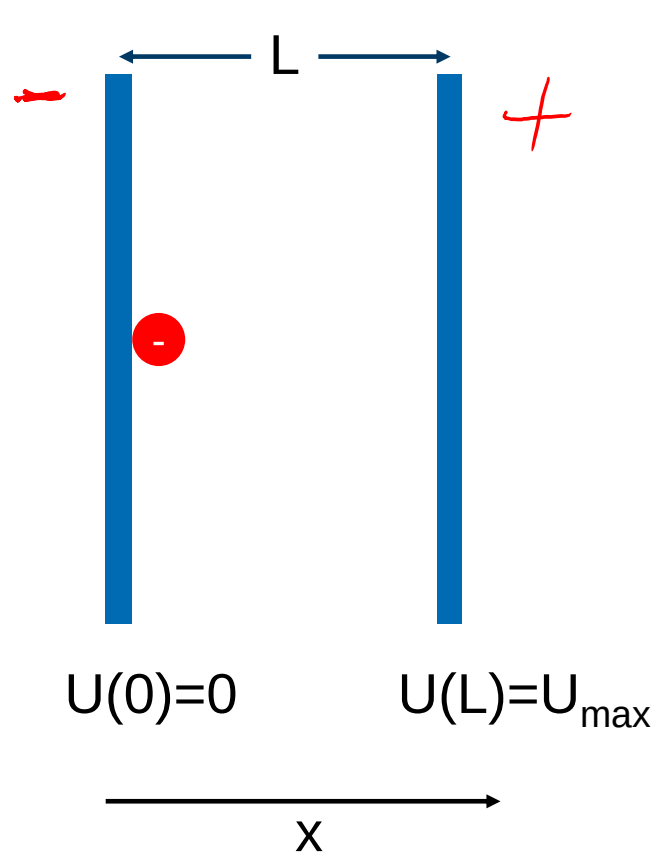


Halliday-Resnick-Walker (2003) Abb. 25.9

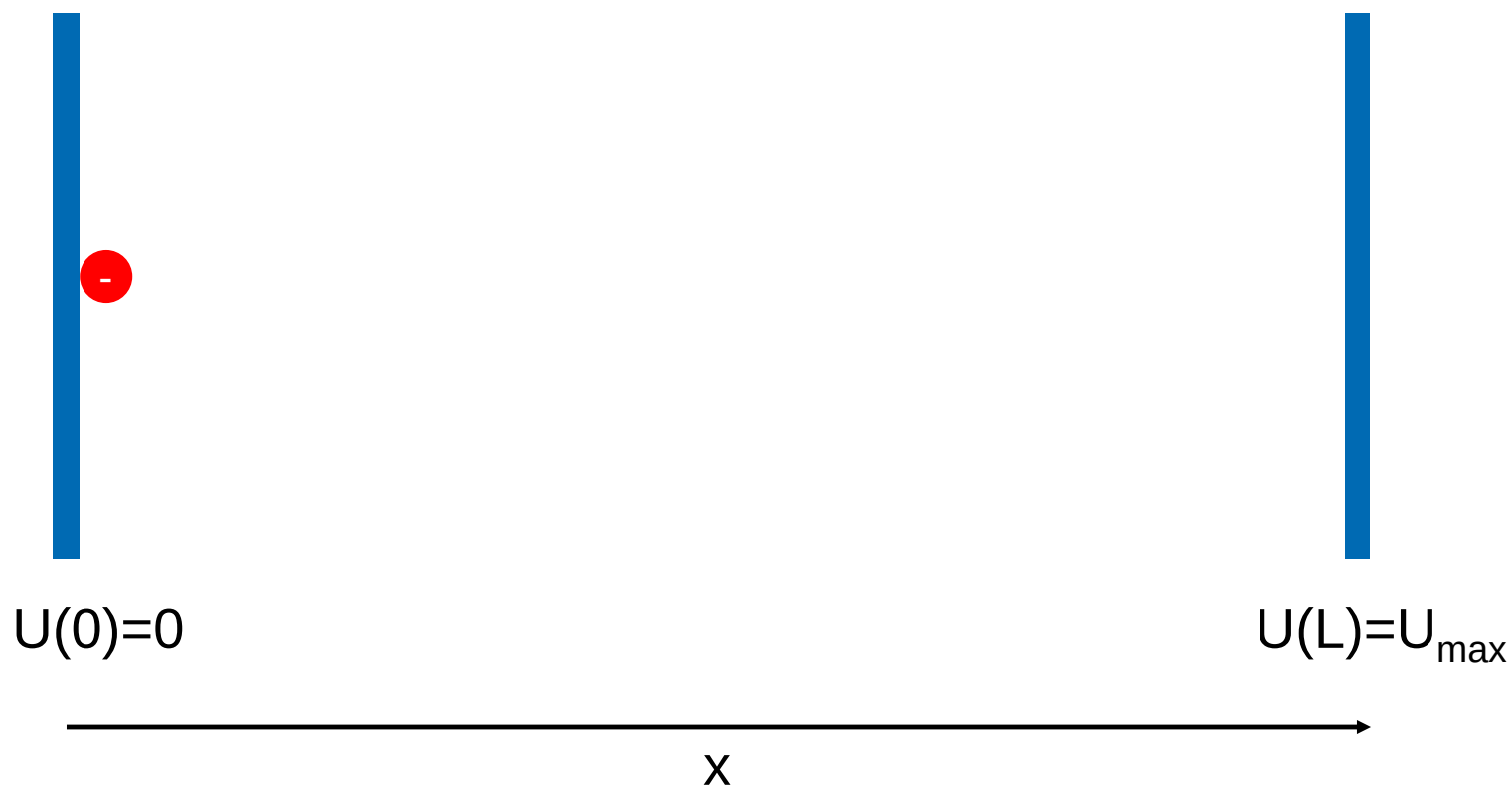
Planparallele Platten:



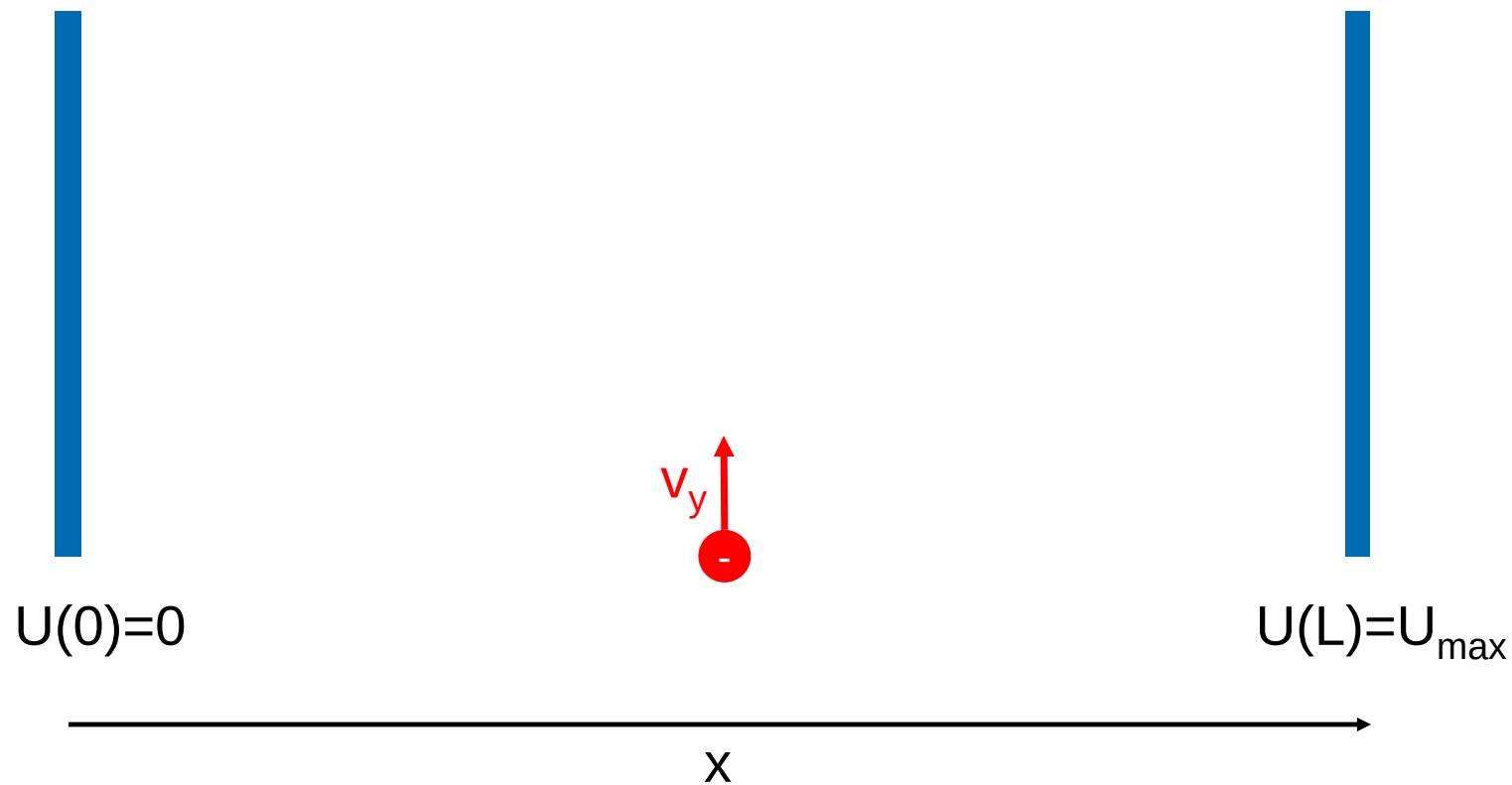
Elektron im Plattenkondensator:



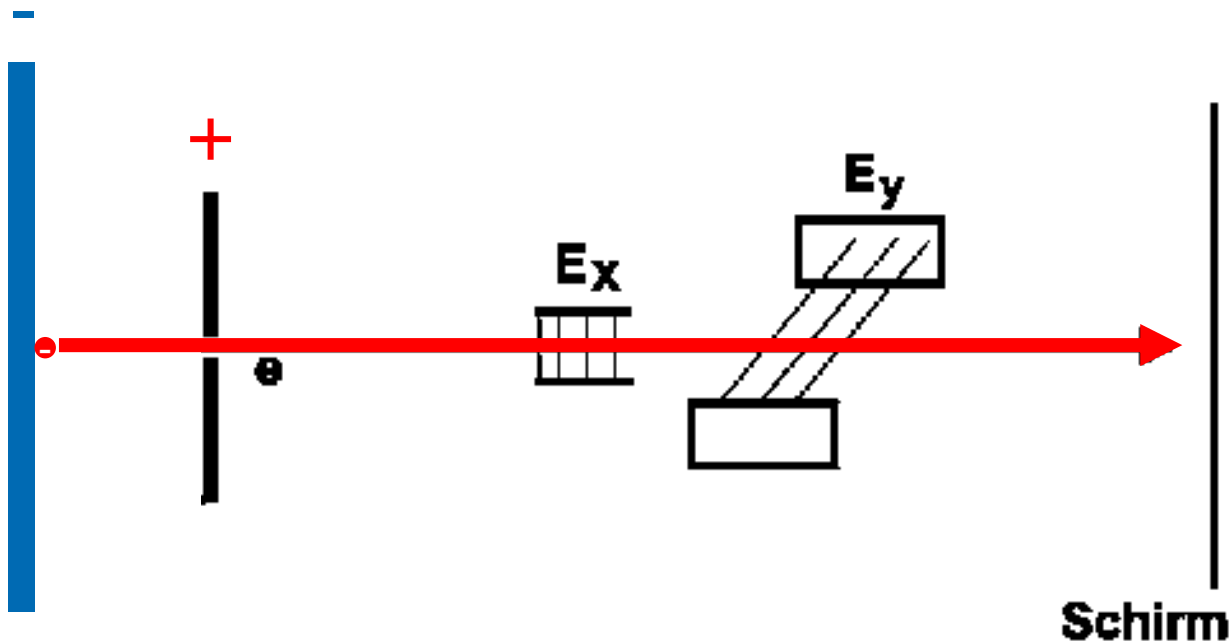
Beschleunigung von Elektronen



Beschleunigung von Elektronen



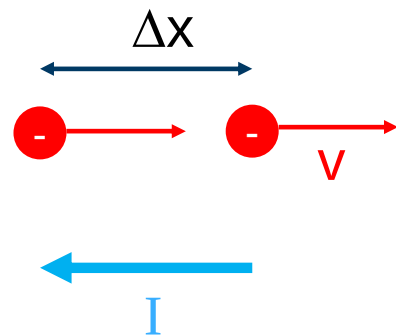
(Analoges) Oszilloskop



www.ebay.de

Skript, Experimentalphysik III, Uni Tübingen, S. 53

Elektrischer Strom



Elektrischer Strom



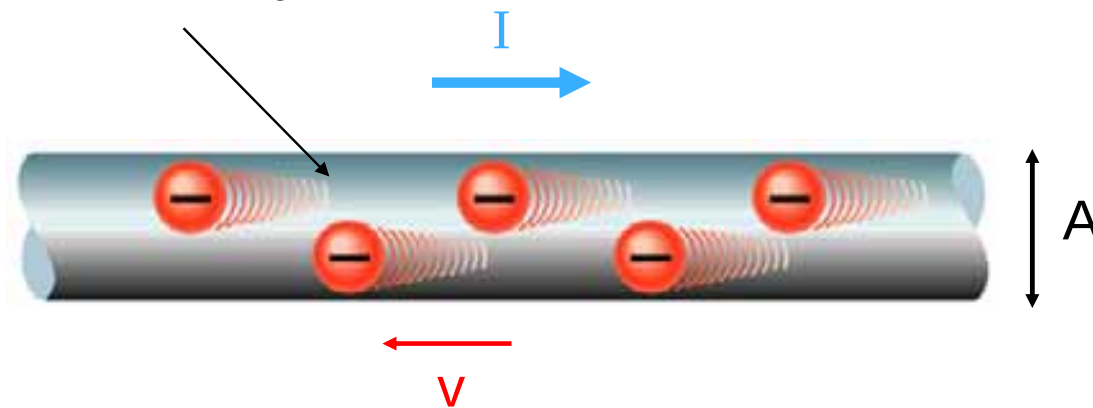
wikipedia.de

Elektrischer Strom

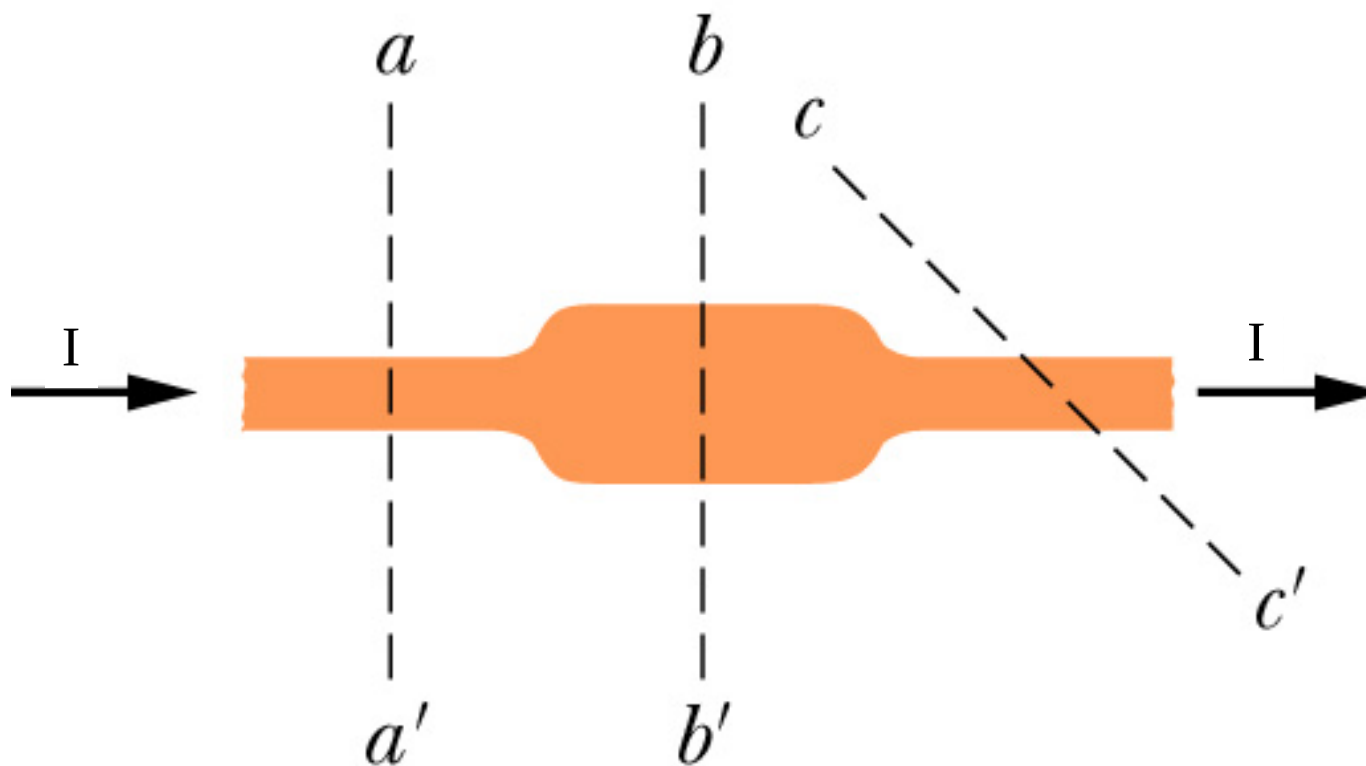


www.mswerklehrmittel.de

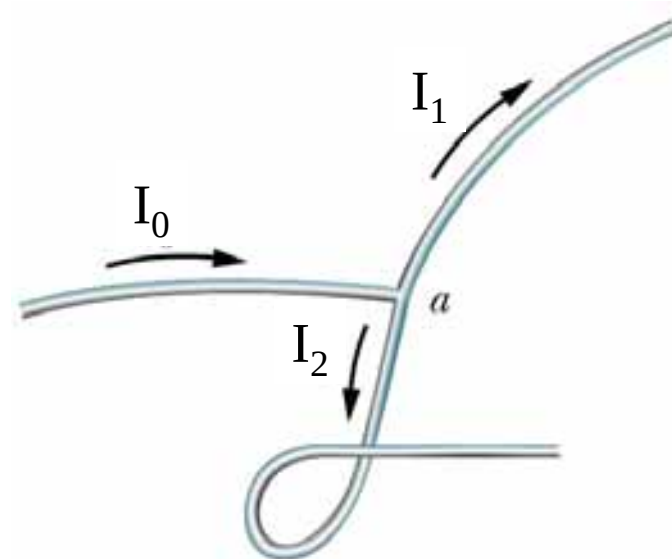
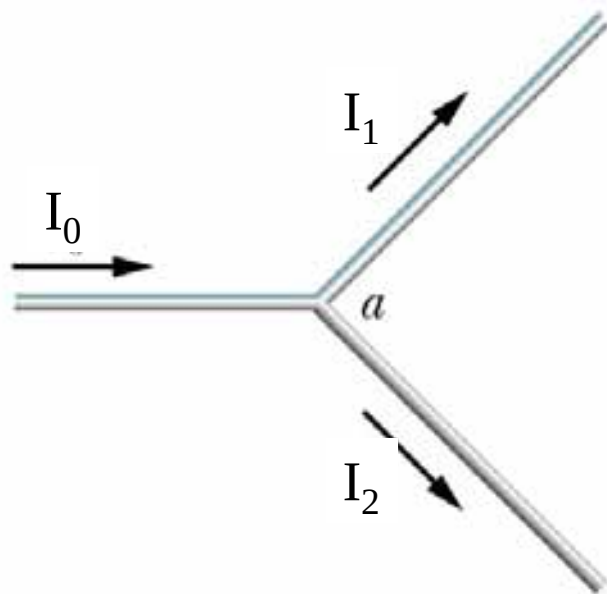
Elektronendichte n_e



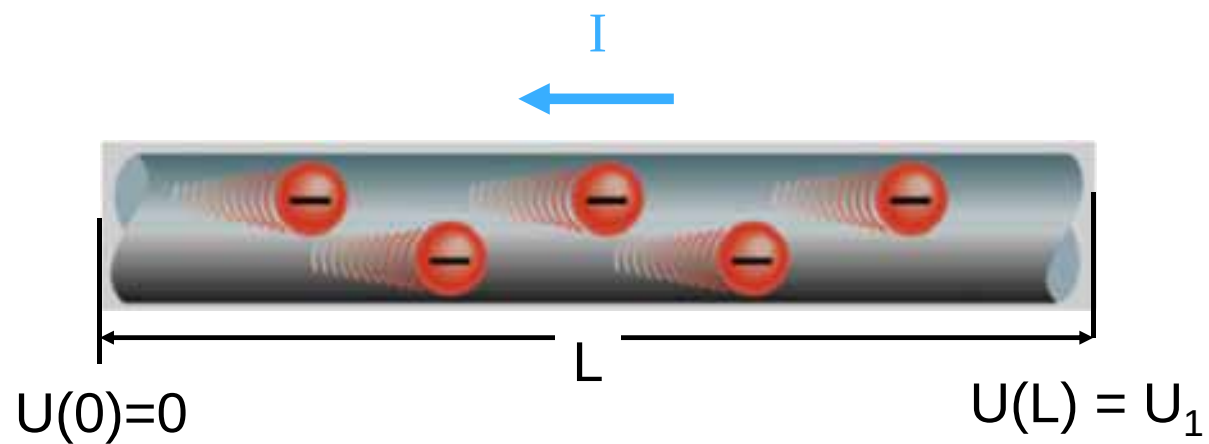
Halliday-Resnick-Walker (2003) Abb. 27.2



Halliday-Resnick-Walker (2003) Abb. 27.2

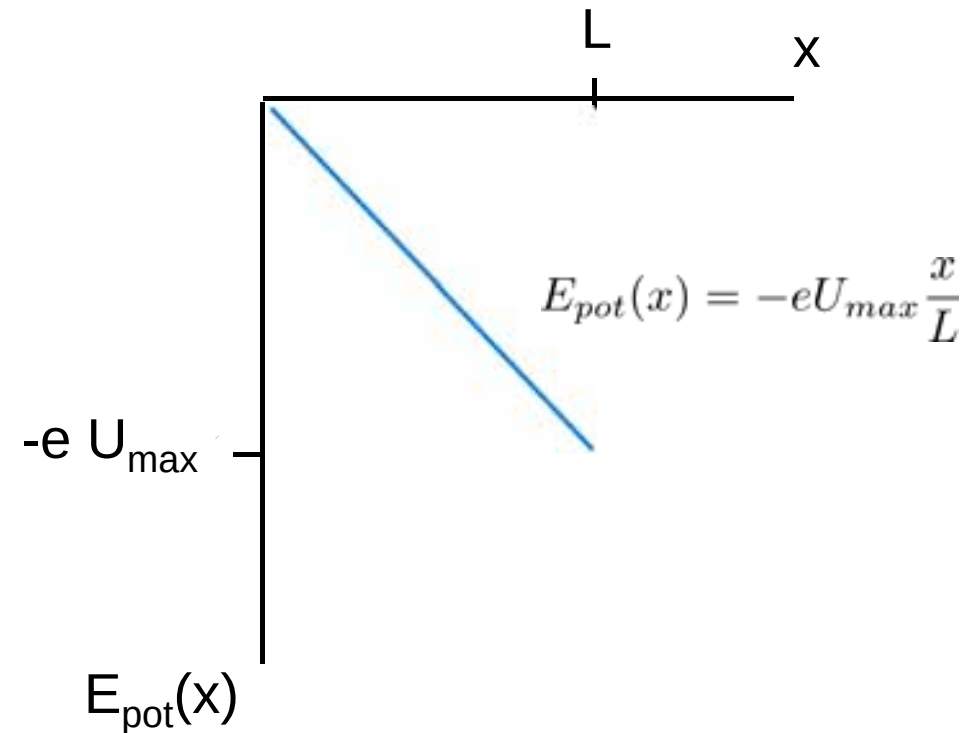
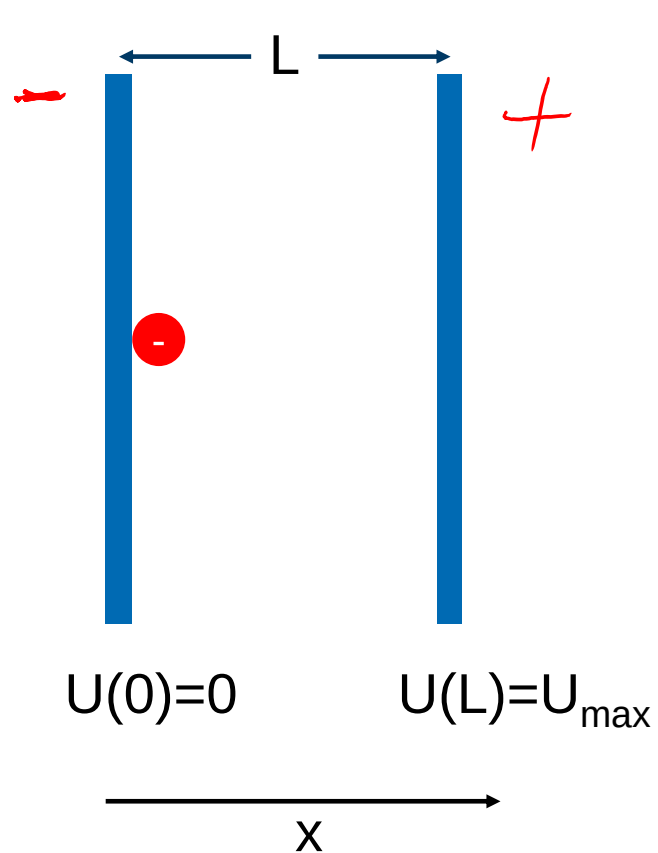


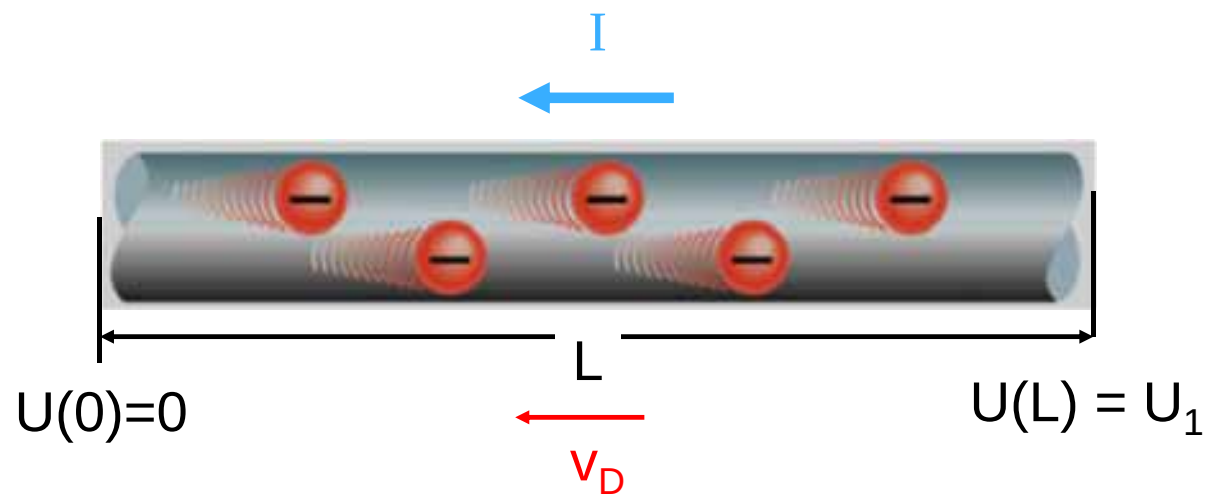
Halliday-Resnick-Walker (2003) Abb. 27.3



Halliday-Resnick-Walker (2003) Abb. 27.2

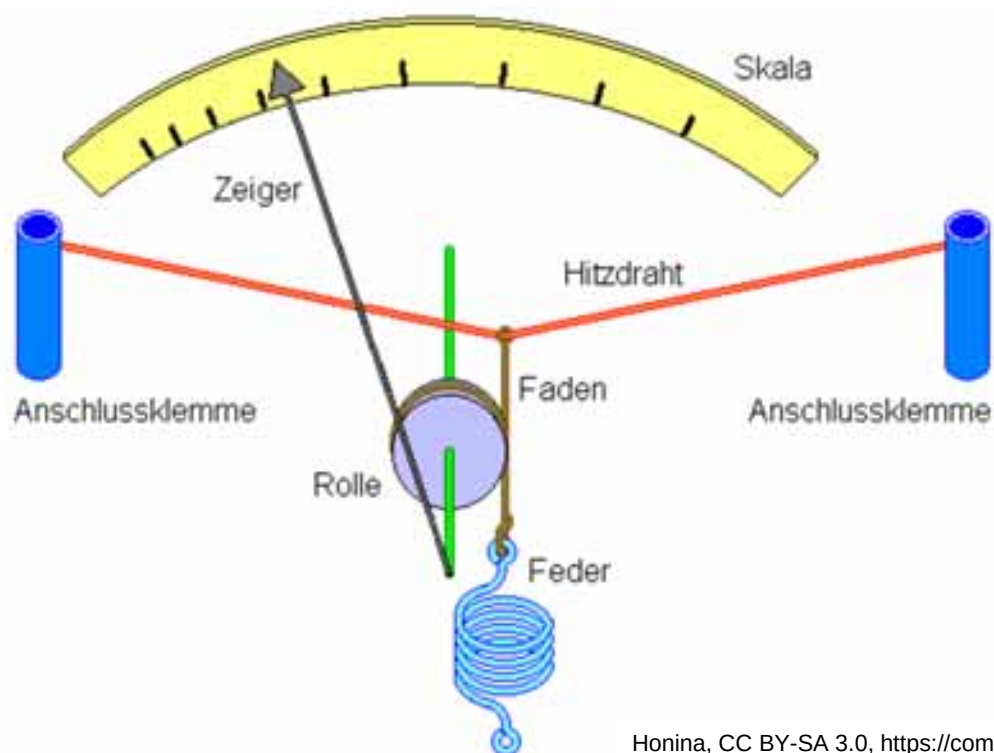
Elektron im Plattenkondensator:





Halliday-Resnick-Walker (2003) Abb. 27.2

Hitzdraht-Amperemeter



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Erzeugung von Elektronen

- Feldemission/Gasentladung

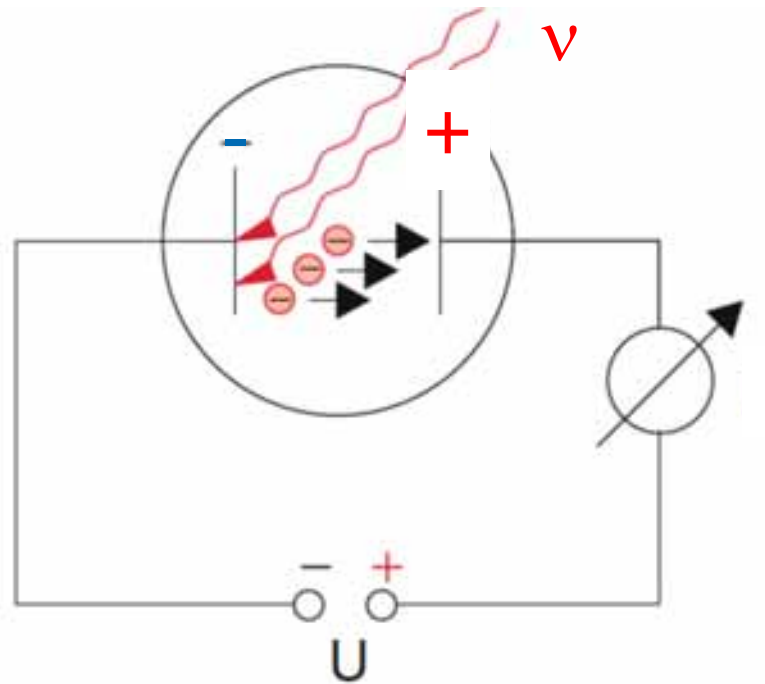


Erzeugung von Elektronen

■ Fotoeffekt



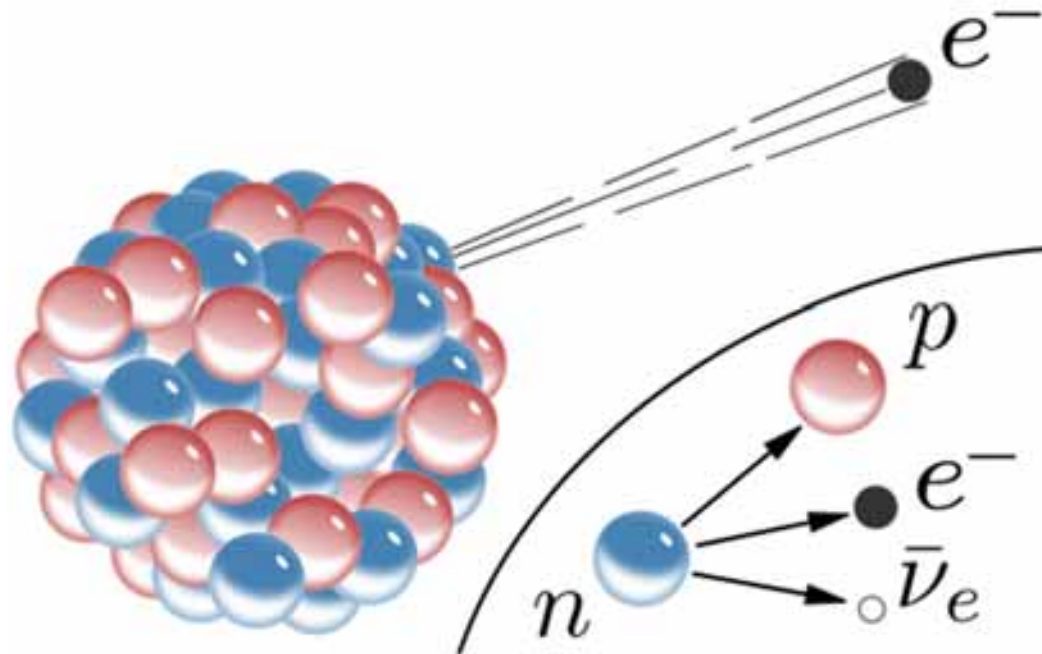
P. Lenard



Demtröder. Abb. 3.9

Erzeugung von Elektronen

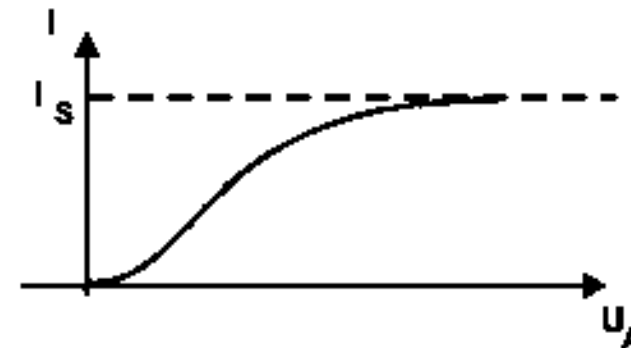
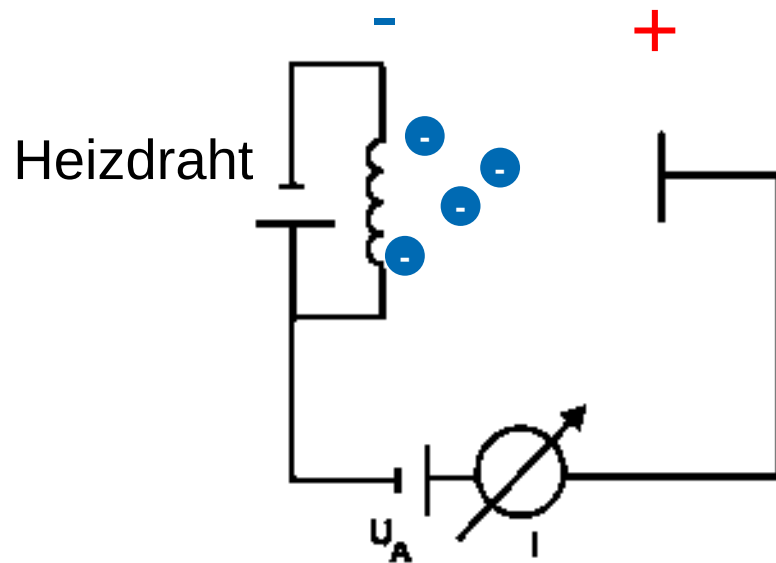
■ β - Strahlung



https://commons.wikimedia.org/wiki/File:Beta-minus_Decay.svg

Erzeugung von Elektronen

■ Glühemission



Skript, Experimentalphysik III, Uni Tübingen, S. 51